

Yusuf Badar

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EDUCATION

Stony Brook University

Stony Brook, NY

B.S. in Computer Science; B.S. in Applied Mathematics

Expected Dec 2026

- Computer Science Honors Program; Sigma Chi Chapter Founder
- Clubs: Quant Club, Competitive Programming Club, Computing Society, Math Club
- Relevant Coursework: System Fundamentals, Database Systems, Software Development, Theory of Computation, Computational Geometry, FinMath, Probability, Combinatorics & Graph Theory, Data Structures & Algorithms

EXPERIENCE

Software Engineer Intern

May 2026 – Aug 2026

Visa

Foster City, CA

- Built and deployed an agentic AI service end-to-end (Angular frontend, FastAPI backend, standalone agent microservice on Postgres) that maps technical database columns to governed business terms with source-traced definitions and a human review loop, grounded on a knowledge base of over 17,000 metadata records.
- Trained an XGBoost confidence model over 11 measured features, raising ROC-AUC from 0.55 to 0.80 against the baseline scorer, and used A/B testing to cut generation latency 2x and output token cost 17% during enrichment.
- Onboarded the service and created a CI/CD pipeline with SonarQube, Checkmarx SAST, and Nexus IQ scanning, backed by 516 tests at 81% line coverage.

Technical Analyst

Feb 2025 – May 2026

Fourier Fund

Stony Brook, NY

- Conducted equity research (P/E, EV/EBITDA, EPS) and financial modeling with algorithms within industries.
- Integrated financial, ESG, and regulatory analysis with Monte Carlo simulations to estimate price and risk.
- Led a team of 4 to create a portfolio beating the S&P 500 Index by 18.4% (annual) and designed a vectorized backtesting engine (NumPy/Pandas) with pluggable factors and transaction-cost modeling.

AI Reasoning Architect

May 2024 – Apr 2026

Outlier

New York, NY

- Increased evaluation throughput by ~20% by systematically structuring ratings and feedback for Meta AI output.
- Developed advanced problems across Discrete Math, Graph Theory, Combinatorics, and Multivariable Calculus, plus adversarial visual/math prompts that improved reasoning for Gemini, ChatGPT, and Google AI Studio.

Software Engineer

Oct 2022 – Apr 2024

GT School

New York, NY

- Developed a **Top-40 AI Education App** on iOS, reaching over 32,000 students; designed and optimized internal data structures and resolved high-priority bugs across UI and networking paths which were tested on TestFlight.
- Produced and edited assets using MidJourney, HeyGen/ElevenLabs, and CapCut; contributed to a social brand reaching 20M+ views and 150k subscribers globally.

RESEARCH & PROJECTS

Research Leader

Jan 2025 – Sep 2025

Blockchain Business Lab

Stony Brook, NY

- Led an 8-member research team on crypto and blockchain technology for fine-tuning DeepSeek-R1.
- Designed a lightweight LLM training pipeline on Kaggle using Unsloth, LoRA, and Hugging Face APIs.
- Integrated cryptocurrency data to fine-tune the model for performance, tracked via Weights & Biases.

AI Trading Bot | Python, PyTorch, NumPy/Pandas, Hugging Face, Yahoo Finance API

Jan 2025

- Created a software in Python/PyTorch to ingest Yahoo Finance news, prices, and sentiment for modeling price.
- Employed NLP and neural networks to extract consumer sentiment from HuggingFace and generate predictions.
- Backtesting outperformed the S&P 500 by 13% and used real-time data for present-day automated trades.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript, MIPS Assembly

Web Development: Node.js, React, Next.js, TypeScript, MongoDB, PostgreSQL, Express.js, HTML, CSS

Tools: AWS, Docker, Git, Jira, Linux, Valgrind, Arduino, Bash

Skills: Agile Development, CI/CD, SCRUM, Microservice, Object-Oriented Programming, Unit & Integration Testing